DeepOcean Nexus Global SubSea Communications Infrastructure Platform Documentation

Project Title: DeepOcean Nexus

Case Study: 134 : DeepOcean Nexus – Global Subsea Communications Infrastructure Platform

Student Details:

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- **Subject:** DevOps

Live Link:-

https://deepocean-nexus.onrender.com/

Git-Hub Link:-

https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project

Problem Statement:- DeepOcean Nexus manages critical global submarine communication infrastructure that supports internet connectivity, government communications, and telecommunications worldwide.The existing infrastructure relies on multiple independent systems for deployment, monitoring, logging, and disaster recovery. This creates challenges such as manual operations, delayed incident detection, lack of centralized visibility, slow recovery from failures, and limited scalability.To overcome these challenges, a cloud-native DevOps platform is required that can automate infrastructure provisioning, application deployment, monitoring, logging, alerting, and disaster recovery while ensuring high availability, reliability, and operational efficiency.

Existing Problems:-

1. Manual Deployment Process
2. No Automated Recovery
3. Disaster Recovery Challenges
4. Poor Visibility Of System Status
5. Delayed Incident Detection
6. Service Downtime Risk

Proposed Solution:- The solution uses Docker containers, Kubernetes orchestration, Jenkins CI/CD pipelines, Terraform infrastructure automation, Prometheus monitoring, Grafana dashboards, Fluent Bit logging, and cloud deployment on Render.

This enables:

- Automated application deployment
- Infrastructure as Code (IaC)
- Containerized applications
- Continuous Integration and Continuous Deployment (CI/CD)
- Real-time monitoring
- Centralized logging
- Automated alerts
- Disaster recovery planning
- High availability and resilience
- Improved operational visibility

Technologies Used:-

Git & GitHub – Used for code management and repository hosting.

Docker & Docker Hub – Used for application containerization and image storage.

Jenkins – Used for CI/CD automation and deployment pipelines.

Kubernetes – Used for application deployment, scaling, and management.

Terraform – Used for infrastructure provisioning and automation.

Flask – Used for backend development and API creation.

HTML, CSS & JavaScript – Used for frontend dashboard design and user interaction.

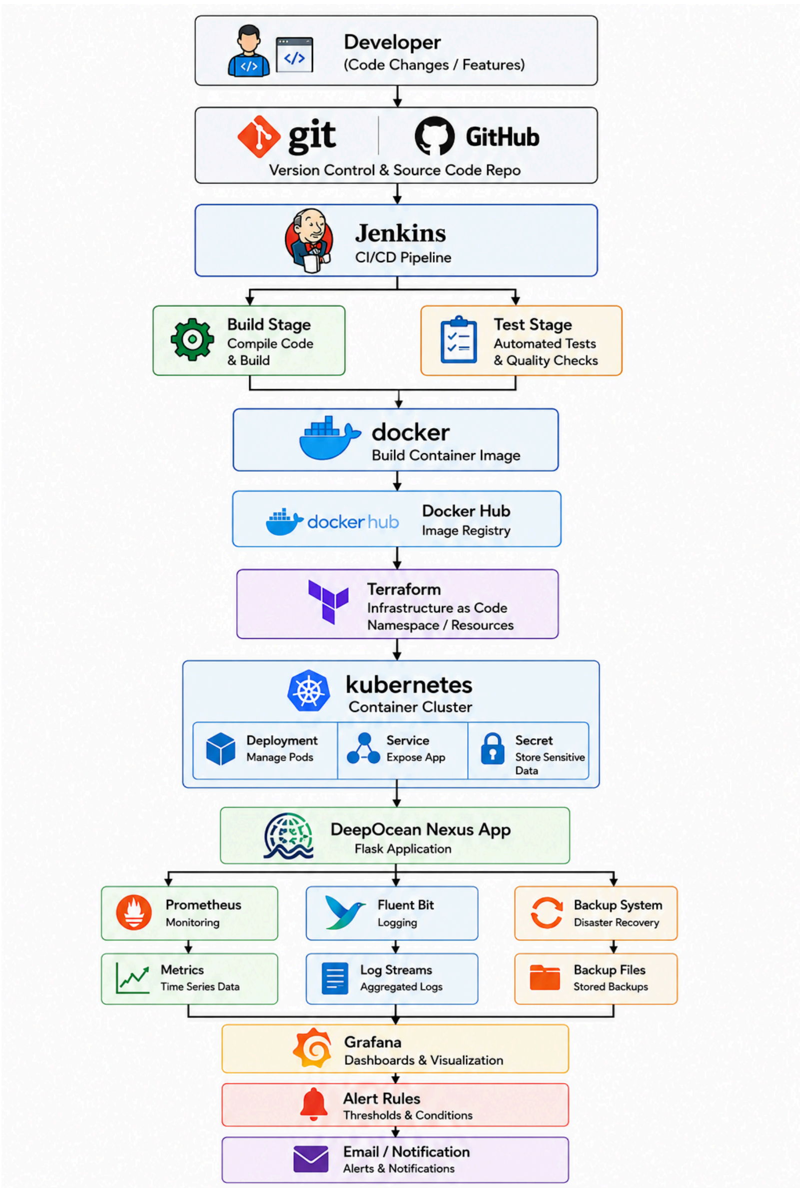
Prometheus – Used for monitoring and metrics collection.

Grafana – Used for dashboard visualization and performance tracking.

Fluent Bit – Used for centralized log collection and management.

Render – Used for cloud hosting and live deployment.

Architecture:-



Executive Summary:- DeepOcean Nexus is a cloud-native DevOps platform designed for monitoring and managing global submarine communication cable infrastructure.

The system provides:

- Real-time cable monitoring
- Infrastructure automation
- Continuous deployment
- Centralized logging
- Alert management
- Disaster recovery planning
- Operational visibility

The platform uses modern DevOps technologies such as Docker, Kubernetes, Jenkins, Terraform, Prometheus, Grafana, Fluent Bit, and Render Cloud.

Folder Structure:-

```
DeepOcean_Nexus
│
├── app
│   ├── templates
```

```
│   ├── static
│   ├── app.py
│   ├── fake_telemetry.py
│   ├── requirements.txt
│
├── jenkins
│
├── kubernetes
│
├── terraform
│
├── monitoring
│
├── logging
│
├── backup
│
└── README.md
```

Working Flow:-

Step 1 :- Developer writes code.

Step 2 :- Code pushed to GitHub.

Step 3 :- Jenkins CI/CD starts.

Benefits:

- Faster releases
- Reduced errors
- Automation

Step 4 :- Docker image created and pushed.

Purpose:

- Containerized application
- Environment consistency
- Easy deployment

Step 5 :- Terraform provisions infrastructure.

Commands:

- terraform init
- terraform plan
- terraform apply

Benefits:

- Automated infrastructure
- Version controlled infrastructure
- Repeatable deployment

Step 6 :- Kubernetes deploys application.

Resources Used:

- Deployment
- Service
- Secret

Features:

- Self-healing
- Auto-restart
- Scalability

Step 7 :- Prometheus monitors metrics.

Metrics collected:

- Active cables
- Throughput
- Security score
- Packet loss

Endpoint: /metrics

Step 8 :- Grafana visualizes metrics.

Dashboard Features:

- Live monitoring
- Throughput tracking
- Alert visibility
- Network health monitoring

Step 9 :- Alerts generated.

Alert Conditions:

- High latency
- Packet loss
- Security incidents
- Service downtime

Step 10 :- Backup and recovery available.

Failure Scenarios:

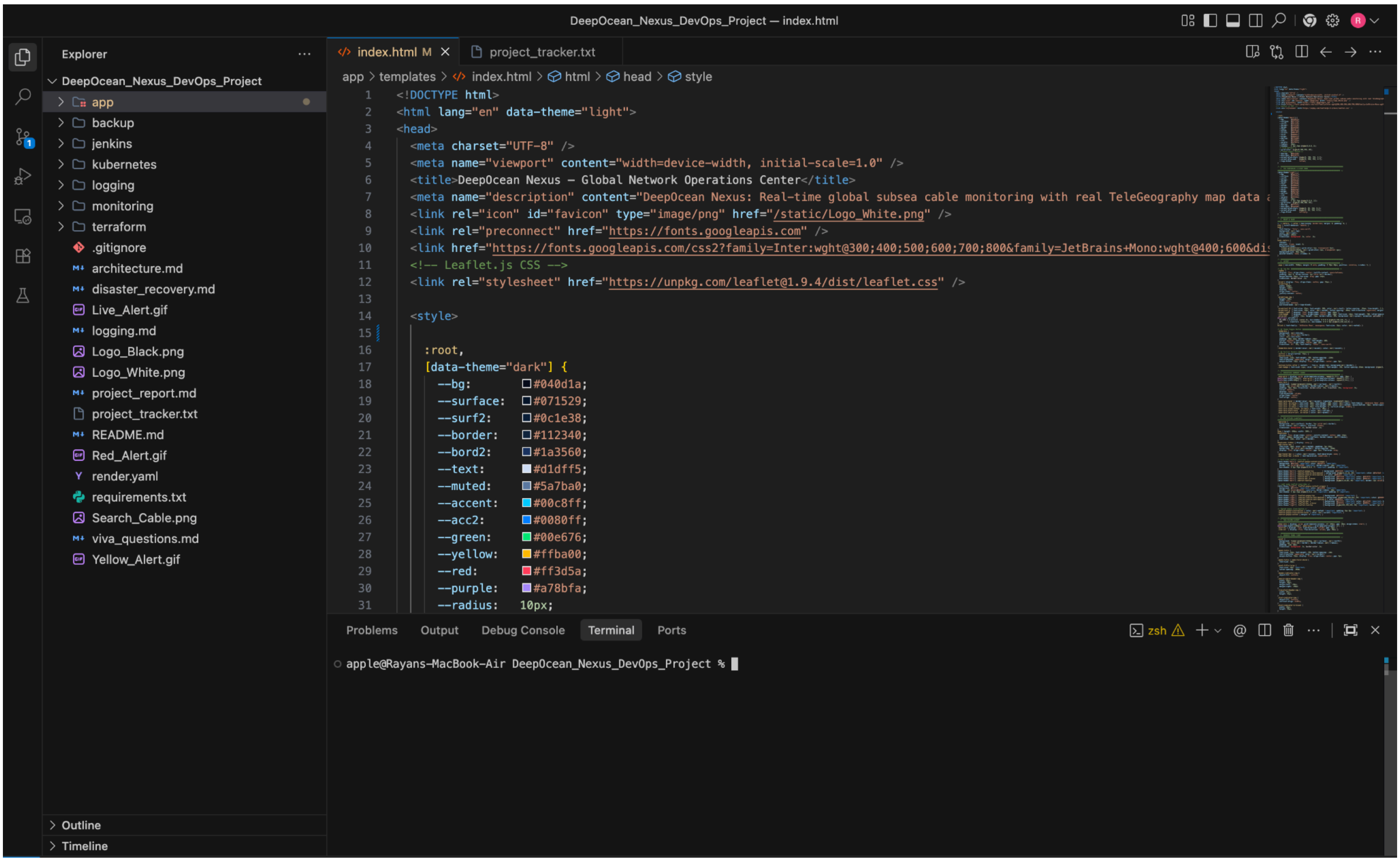
- Application crash
- Container failure
- Node failure
- Data corruption

Recovery Process:

[Backup] → [Restore] → [Validation] → [Service Recovery]

Working Flow With Images:-

Step 1 :- Developer writes code.



Step 2 :- Code pushed to GitHub.

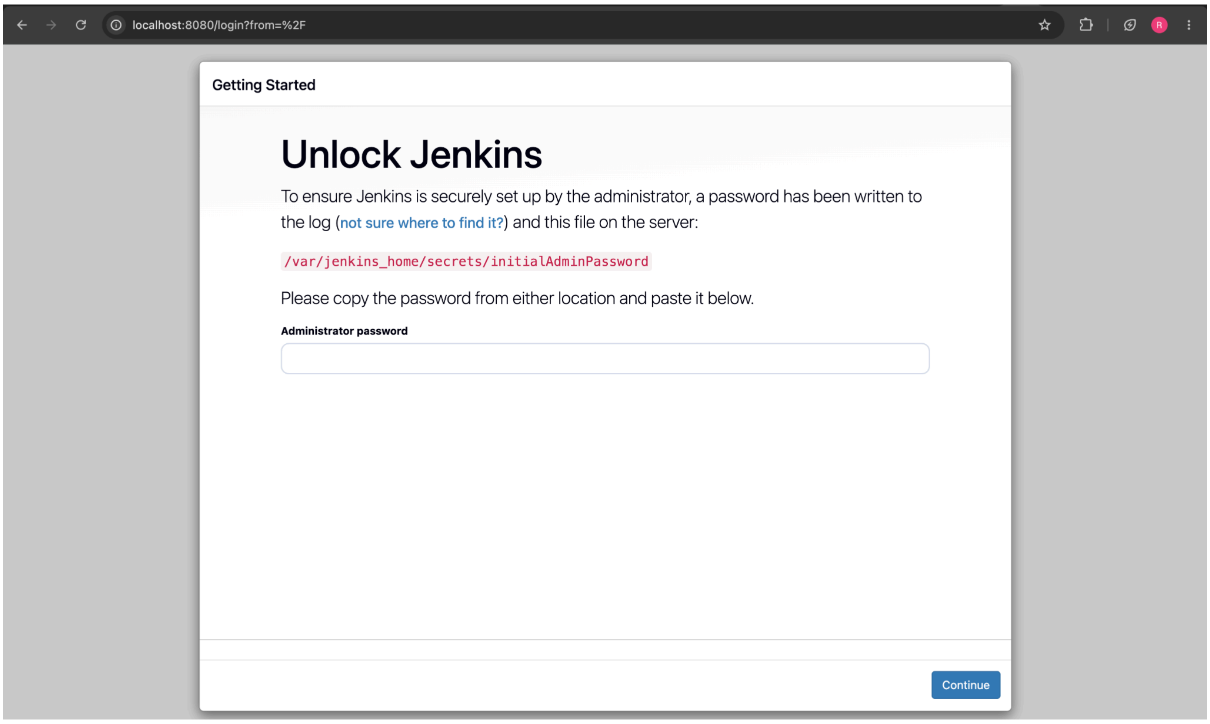
```
apple@Rayans-MacBook-Air ~ % cd ~/Desktop/DeepOcean_Nexus_DevOps_Project
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git init
Initialized empty Git repository in /Users/apple/Desktop/DeepOcean_Nexus_DevOps_Project/.git/
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        README.md
        app/
        architecture.md
        disaster_recovery.md
        jenkins/
        kubernetes/
        logging.md
        monitoring/
        project_report.md
        project_tracker.txt
        terraform/
        viva_questions.md

nothing added to commit but untracked files present (use "git add" to track)
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git add .
```

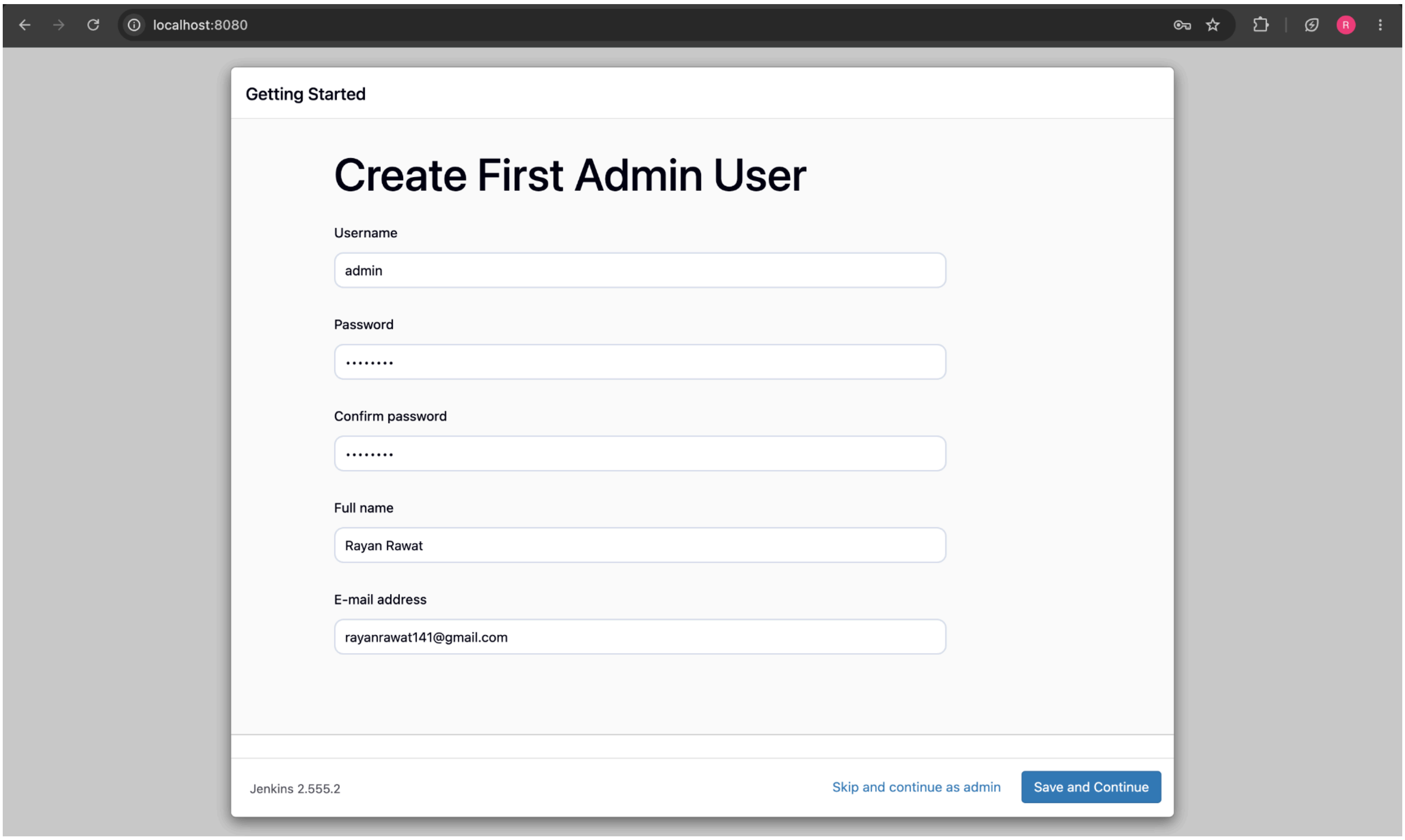
```
[apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   README.md
        new file:   app/Dockerfile
        new file:   app/app.py
        new file:   app/requirements.txt
        new file:   app/templates/index.html
        new file:   architecture.md
        new file:   disaster_recovery.md
        new file:   jenkins/Jenkinsfile
        new file:   kubernetes/deployment.yaml
        new file:   kubernetes/secret.yaml
        new file:   kubernetes/service.yaml
        new file:   logging.md
        new file:   monitoring/docker-compose.yml
        new file:   monitoring/prometheus.yml
        new file:   project_report.md
        new file:   project_tracker.txt
        new file:   terraform/.terraform.lock.hcl
        new file:   terraform/main.tf
        new file:   terraform/terraform.tfstate
        new file:   viva_questions.md
```

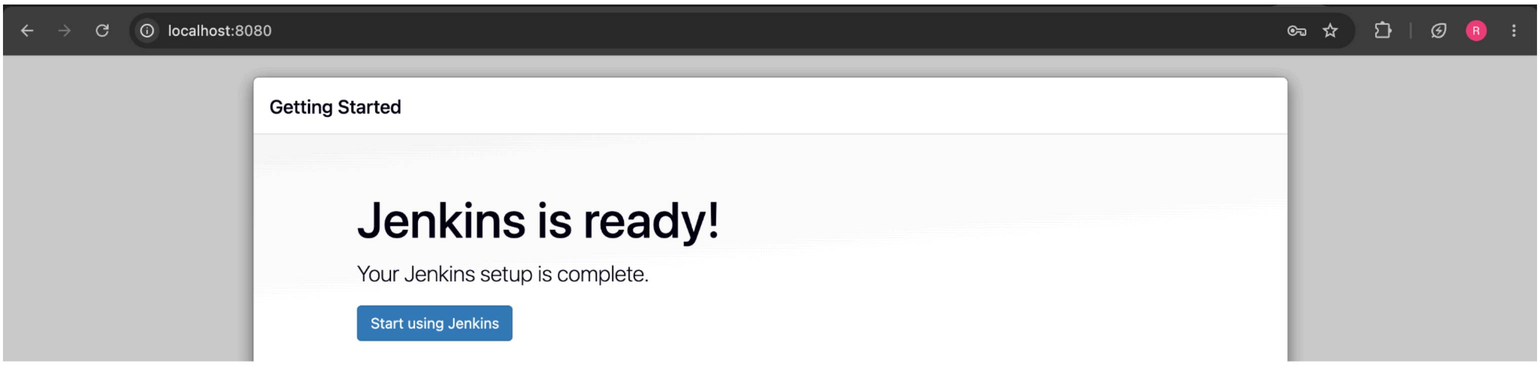
```
[apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git commit -m "Initial project upload"
[main (root-commit) 32986cd] Initial project upload
 21 files changed, 1264 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 app/Dockerfile
 create mode 100644 app/app.py
 create mode 100644 app/requirements.txt
 create mode 100644 app/templates/index.html
 create mode 100644 architecture.md
 create mode 100644 disaster_recovery.md
 create mode 100644 jenkins/Jenkinsfile
 create mode 100644 kubernetes/deployment.yaml
 create mode 100644 kubernetes/secret.yaml
 create mode 100644 kubernetes/service.yaml
 create mode 100644 logging.md
 create mode 100644 monitoring/docker-compose.yml
 create mode 100644 monitoring/prometheus.yml
 create mode 100644 project_report.md
 create mode 100644 project_tracker.txt
 create mode 100644 terraform/.terraform.lock.hcl
 create mode 100644 terraform/main.tf
 create mode 100644 terraform/terraform.tfstate
 create mode 100644 viva_questions.md
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
```

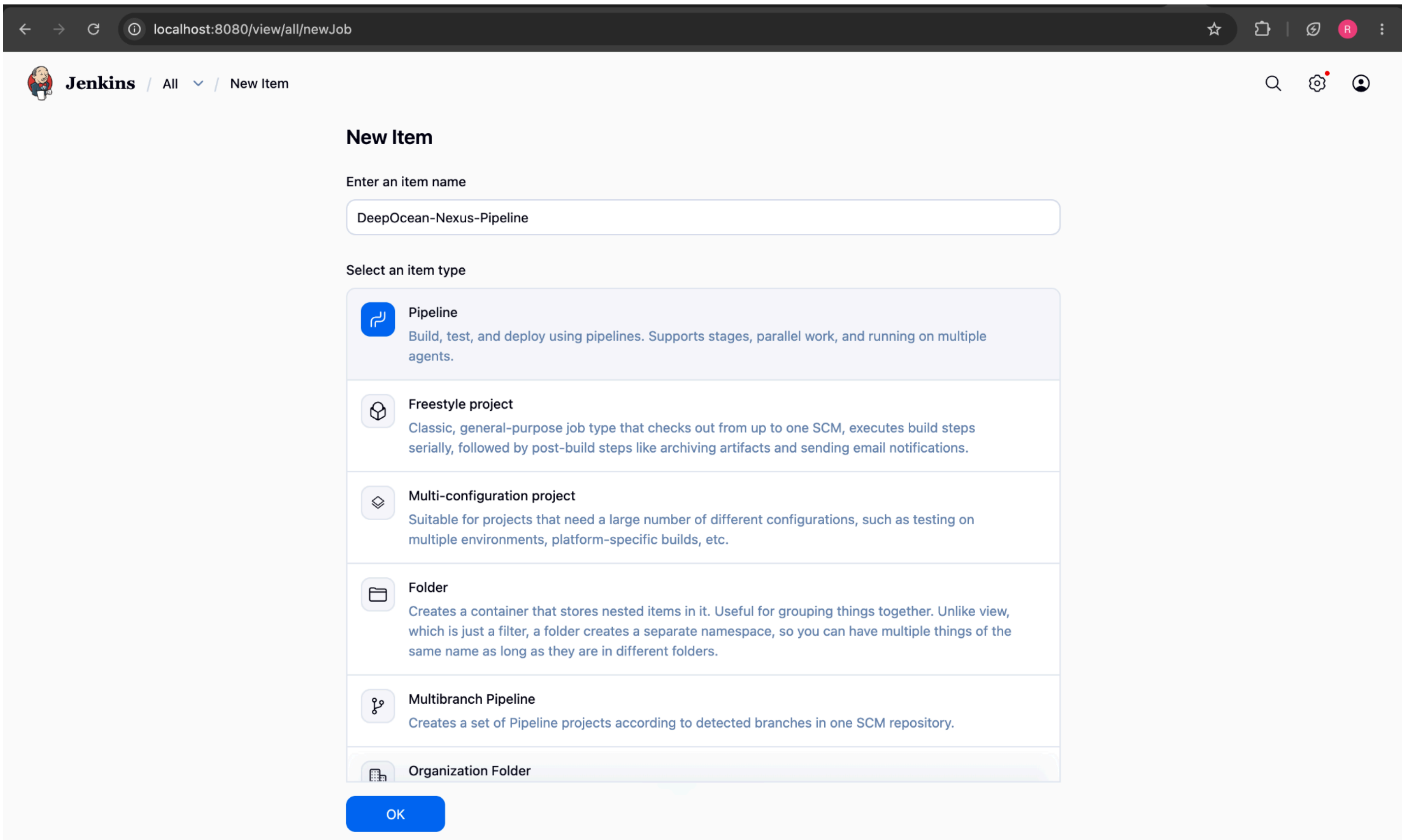
```
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git remote add origin https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project.git
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git remote -v
origin  https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project.git (fetch)
origin  https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project.git (push)
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % git push -u origin main
Enumerating objects: 29, done.
Counting objects: 100% (29/29), done.
Delta compression using up to 8 threads
Compressing objects: 100% (26/26), done.
Writing objects: 100% (29/29), 15.37 KiB | 5.12 MiB/s, done.
Total 29 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
```

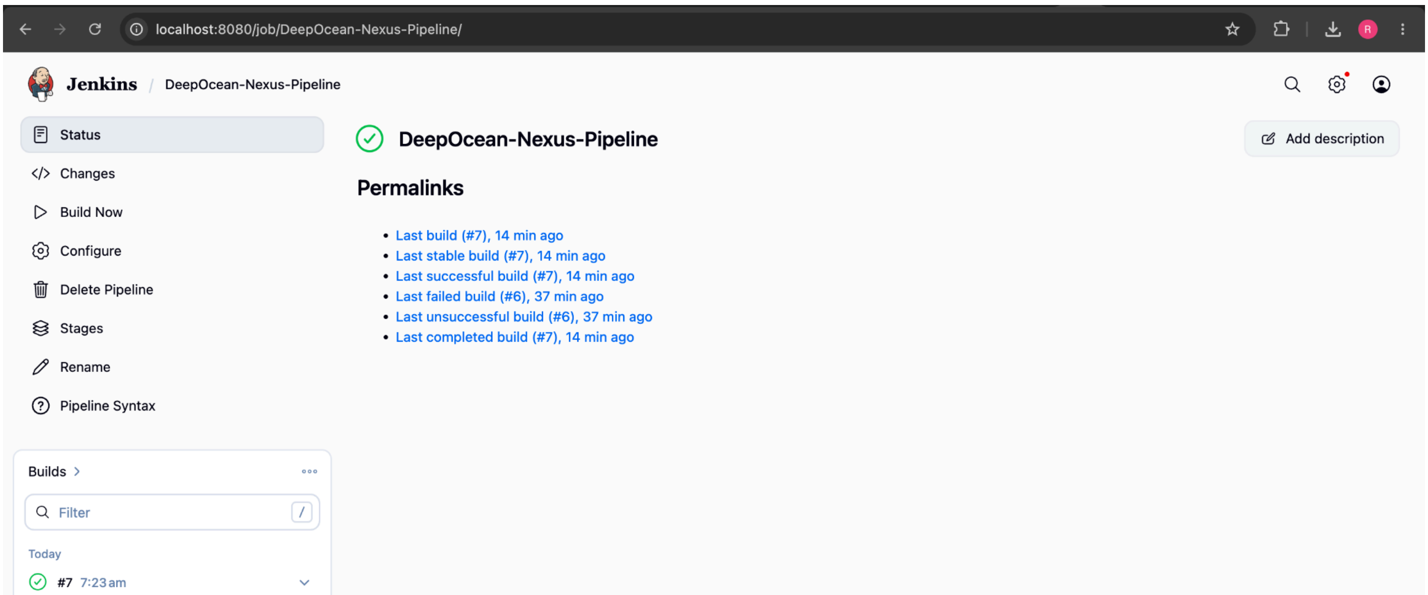
Step 3 :- Jenkins CI/CD starts.

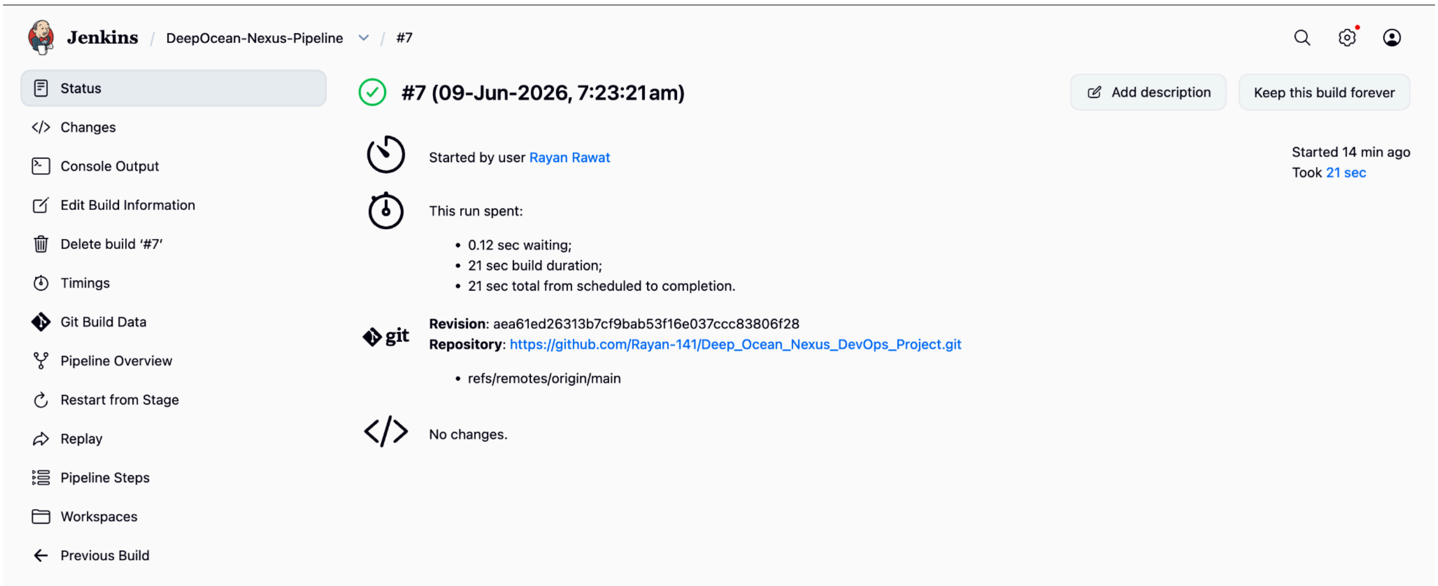


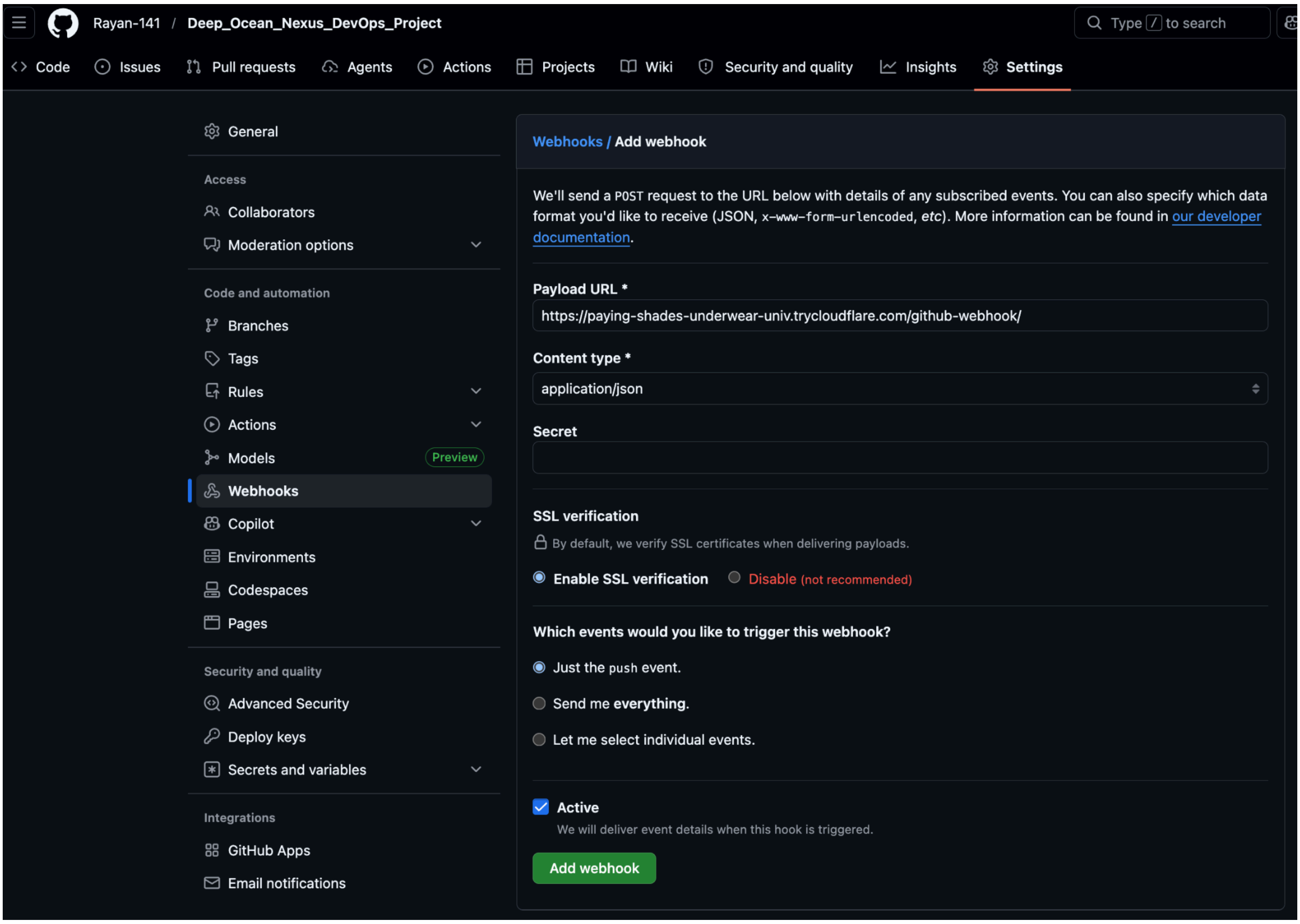

localhost:8080
Getting Started
Create First Admin User
Username
admin
Password
Confirm password
Full name
Rayan Rawat
E-mail address
rayanrawat141@gmail.com
Jenkins 2.555.2
Skip and continue as admin
Save and Continue

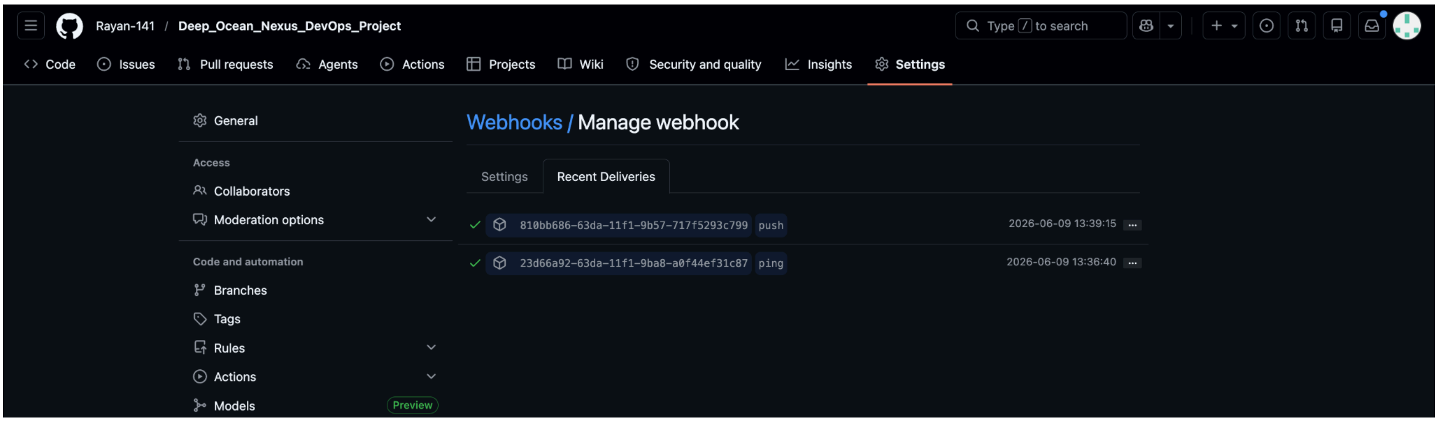

localhost:8080
Getting Started
Jenkins is ready!
Your Jenkins setup is complete.
Start using Jenkins


localhost:8080/view/all/newJob
Jenkins
All
New Item
New Item
Enter an item name
DeepOcean-Nexus-Pipeline
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
OK


localhost:8080/job/DeepOcean-Nexus-Pipeline/
Jenkins
DeepOcean-Nexus-Pipeline
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
DeepOcean-Nexus-Pipeline
Add description
Permalinks
Last build (#7), 14 min ago
Last stable build (#7), 14 min ago
Last successful build (#7), 14 min ago
Last failed build (#6), 37 min ago
Last unsuccessful build (#6), 37 min ago
Last completed build (#7), 14 min ago
Builds
Filter
Today
#7 7:23 am


Jenkins / DeepOcean-Nexus-Pipeline / #7
Status
Changes
Console Output
Edit Build Information
Delete build '#7'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#7 (09-Jun-2026, 7:23:21am)
Add description
Keep this build forever
Started by user Rayan Rawat
Started 14 min ago
Took 21 sec
This run spent:
0.12 sec waiting;
21 sec build duration;
21 sec total from scheduled to completion.
git
Revision: aea61ed26313b7cf9bab53f16e037ccc83806f28
Repository: https://github.com/Rayan-141/Deep_Ocean_Nexus_DevOps_Project.git
refs/remotes/origin/main
No changes.


Rayan-141 / Deep_Ocean_Nexus_DevOps_Project
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
Pages
Security and quality
Advanced Security
Deploy keys
Secrets and variables
Integrations
GitHub Apps
Email notifications
Webhooks / Add webhook
We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in our developer documentation.
Payload URL *
https://paying-shades-underwear-univ.trycloudflare.com/github-webhook/
Content type *
application/json
Secret
SSL verification
By default, we verify SSL certificates when delivering payloads.
Enable SSL verification
Disable (not recommended)
Which events would you like to trigger this webhook?
Just the push event.
Send me everything.
Let me select individual events.
Active
We will deliver event details when this hook is triggered.
Add webhook



Step 4 :- Docker image created and pushed.

```
apple@Rayans-MacBook-Air app % docker build -t rayan221006/deepocean-app:latest .
[+] Building 49.6s (11/11) FINISHED                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                         0.2s
 => => transferring dockerfile: 207B                                                                         0.1s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                          3.5s
 => [auth] library/python:pull token for registry-1.docker.io                                                0.0s
 => [internal] load .dockerignore                                                                            0.1s
 => => transferring context: 2B                                                                              0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:ef442c44cde6d7aec39ae63dad1ced44fa7290790d68c028686cdd2e31a76a95  23.7s
 => => resolve docker.io/library/python:3.11-slim@sha256:ef442c44cde6d7aec39ae63dad1ced44fa7290790d68c028686cdd2e31a76a95  0.1s
 => => sha256:7f7603e6ebb8a9b94a429dcdb69477fc29b25e033752a99d537ba0d87a455212 250B / 250B                  0.5s
 => => sha256:2e6eaa7747ea36ce09c07e556b94db03b4221eaa799d4d0ca387fb13a46d89f1 1.27MB / 1.27MB              1.7s
 => => sha256:473b26fe033cfb7157bf8eccaaec7fb4c2973e971213128c1a817a6aa0132976 14.32MB / 14.32MB           10.7s
 => => sha256:a25cd16f2d8653f652f8292b34b21bfbabdc85d6b39861a24b85f0896df1a95e 30.15MB / 30.15MB           16.5s
 => => extracting sha256:a25cd16f2d8653f652f8292b34b21bfbabdc85d6b39861a24b85f0896df1a95e                   2.8s
 => => extracting sha256:2e6eaa7747ea36ce09c07e556b94db03b4221eaa799d4d0ca387fb13a46d89f1                   0.5s
 => => extracting sha256:473b26fe033cfb7157bf8eccaaec7fb4c2973e971213128c1a817a6aa0132976                   3.1s
 => => extracting sha256:7f7603e6ebb8a9b94a429dcdb69477fc29b25e033752a99d537ba0d87a455212                   0.1s
 => [internal] load build context                                                                            0.6s
 => => transferring context: 3.55MB                                                                          0.5s
 => [2/5] WORKDIR /app                                                                                       1.0s
 => [3/5] COPY requirements.txt .                                                                            0.1s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                17.3s
 => [5/5] COPY . .                                                                                           0.2s
 => exporting to image                                                                                       3.3s
 => => exporting layers                                                                                      2.8s
 => => exporting manifest sha256:946399d57af4d29ef2f8e83ae2f94aeeab088b2c27d18fd82a4d759b23763a6b            0.0s
 => => exporting config sha256:84a5a337ccac9d99e8e640f61f69ec46b2d60370e1e7d3138a7ff732047745fd              0.0s
 => => exporting attestation manifest sha256:ed2d632be6adf68ab03291b38685f8af351d03f126b774bb59dcfcb187baa678  0.0s
 => => exporting manifest list sha256:77ad8783786d76055b2f8bc97489bd3fa8b2b6bae72ae69665e0d7c182e5be42       0.0s
 => => naming to docker.io/rayan221006/deepocean-app:latest                                                  0.0s
 => => unpacking to docker.io/rayan221006/deepocean-app:latest                                               0.4s
```

```
apple@Rayans-MacBook-Air app % docker push rayan221006/deepocean-app:latest
The push refers to repository [docker.io/rayan221006/deepocean-app]
4b9da6413dc1: Pushed
7f7603e6ebb8: Pushed
2acdd9574587: Pushed
77bf51ffb9b5: Pushed
2e6eaa7747ea: Pushed
e9d3934b1544: Pushed
394d5f459c3d: Pushed
473b26fe033c: Pushed
a25cd16f2d86: Pushed
latest: digest: sha256:77ad8783786d76055b2f8bc97489bd3fa8b2b6bae72ae69665e0d7c182e5be42 size: 856
apple@Rayans-MacBook-Air app %
```

Step 5 :- Terraform provisions infrastructure.

```
[apple@Rayans-MacBook-Air app % cd ../terraform
[apple@Rayans-MacBook-Air terraform % terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/kubernetes from the dependency lock file
- Using previously-installed hashicorp/kubernetes v3.2.0

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/kubernetes
- Using previously-installed hashicorp/kubernetes v3.2.0


Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
[apple@Rayans-MacBook-Air terraform % terraform apply -auto-approve
kubernetes_namespace.deepocean: Refreshing state... [id=deepocean]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # kubernetes_namespace.deepocean will be created
  + resource "kubernetes_namespace" "deepocean" {
      + id                              = (known after apply)
      + wait_for_default_service_account = false

      + metadata {
          + generation       = (known after apply)
          + name             = "deepocean"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.
kubernetes_namespace.deepocean: Creating...
kubernetes_namespace.deepocean: Creation complete after 0s [id=deepocean]
│
│ Warning: Deprecated Resource
│
│   with kubernetes_namespace.deepocean,
│   on main.tf line 5, in resource "kubernetes_namespace" "deepocean":
│    5: resource "kubernetes_namespace" "deepocean" {
│
│ Deprecated; use kubernetes_namespace_v1.
│
│ (and 2 more similar warnings elsewhere)
│

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
apple@Rayans-MacBook-Air terraform %
```

```
apple@Rayans-MacBook-Air terraform % kubectl get ns
NAME               STATUS   AGE
deepocean          Active   109s
default            Active   27m
kube-node-lease    Active   27m
kube-public        Active   27m
kube-system        Active   27m
apple@Rayans-MacBook-Air terraform %
```

Step 6 :- Kubernetes deploys application.

```
apple@Rayans-MacBook-Air kubernetes % kubectl apply -f .
deployment.apps/deepocean-app created
secret/deepocean-secret created
service/deepocean-service created
apple@Rayans-MacBook-Air kubernetes %
```

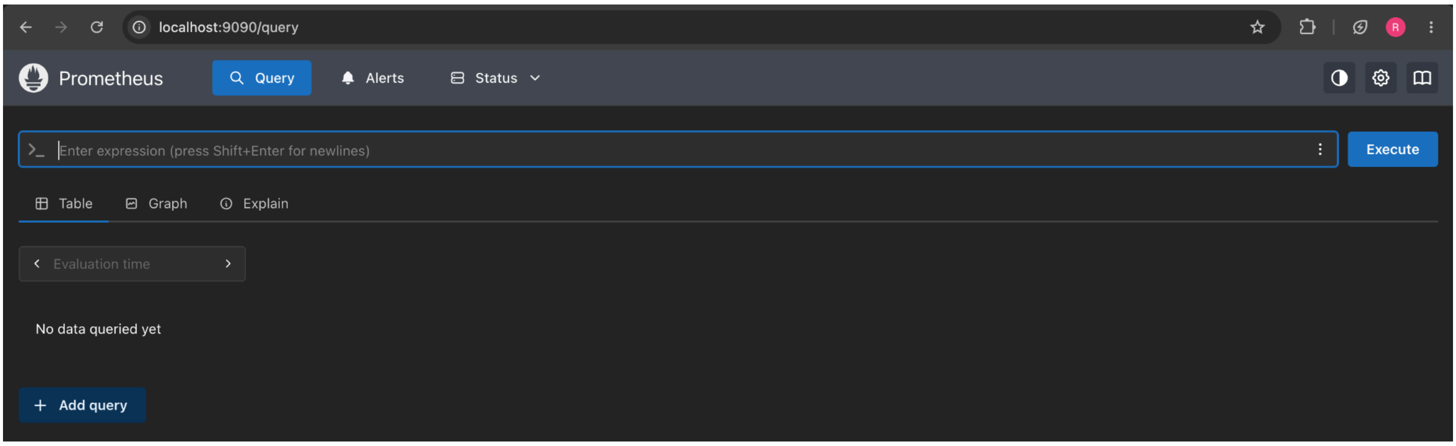
```
apple@Rayans-MacBook-Air kubernetes % kubectl get pods -n deepocean
NAME                             READY   STATUS    RESTARTS   AGE
deepocean-app-5ddfffd555-g5d4f   1/1     Running   0          36s
deepocean-app-5ddfffd555-nn6z5   1/1     Running   0          36s
deepocean-app-5ddfffd555-vkqw9   1/1     Running   0          36s
apple@Rayans-MacBook-Air kubernetes %
```

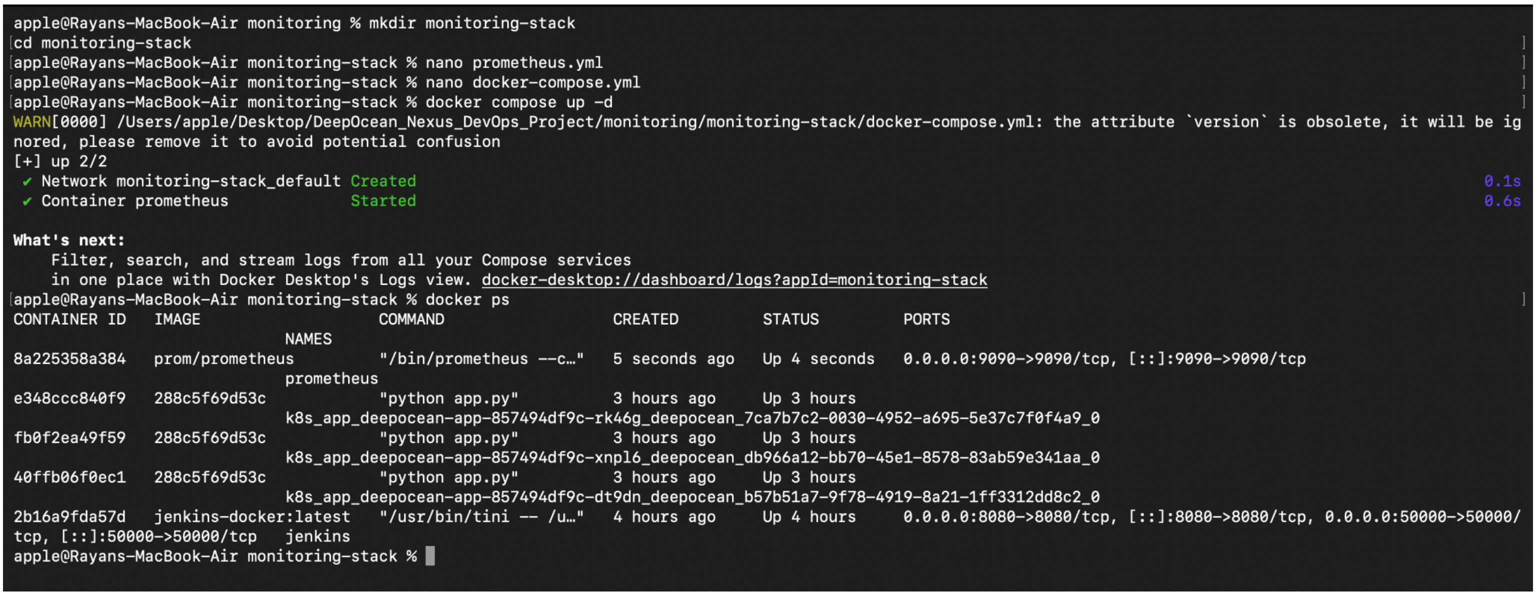
```
apple@Rayans-MacBook-Air kubernetes % kubectl get svc -n deepocean
NAME                TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
deepocean-service   NodePort   10.105.165.192   <none>        5000:30000/TCP   114s
apple@Rayans-MacBook-Air kubernetes %
```

```
apple@Rayans-MacBook-Air kubernetes % kubectl port-forward service/deepocean-service 5000:5000 -n deepocean
Forwarding from 127.0.0.1:5000 -> 5000
Forwarding from [::1]:5000 -> 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
^C%
apple@Rayans-MacBook-Air kubernetes %
```

```
apple@Rayans-MacBook-Air kubernetes % kubectl get pods -n deepocean -w
NAME                              READY   STATUS    RESTARTS   AGE
deepocean-app-5ddfffd555-g5d4f    1/1     Running   0          6m59s
deepocean-app-5ddfffd555-nn6z5    1/1     Running   0          6m59s
deepocean-app-5ddfffd555-vkqw9    1/1     Running   0          6m59s
deepocean-app-5ddfffd555-g5d4f    1/1     Terminating   0          7m28s
deepocean-app-5ddfffd555-g5d4f    1/1     Terminating   0          7m28s
deepocean-app-5ddfffd555-dz2p7    0/1     Pending       0          0s
deepocean-app-5ddfffd555-dz2p7    0/1     Pending       0          0s
deepocean-app-5ddfffd555-dz2p7    0/1     ContainerCreating   0          0s
deepocean-app-5ddfffd555-dz2p7    0/1     Running             0          5s
deepocean-app-5ddfffd555-dz2p7    1/1     Running             0          16s
```

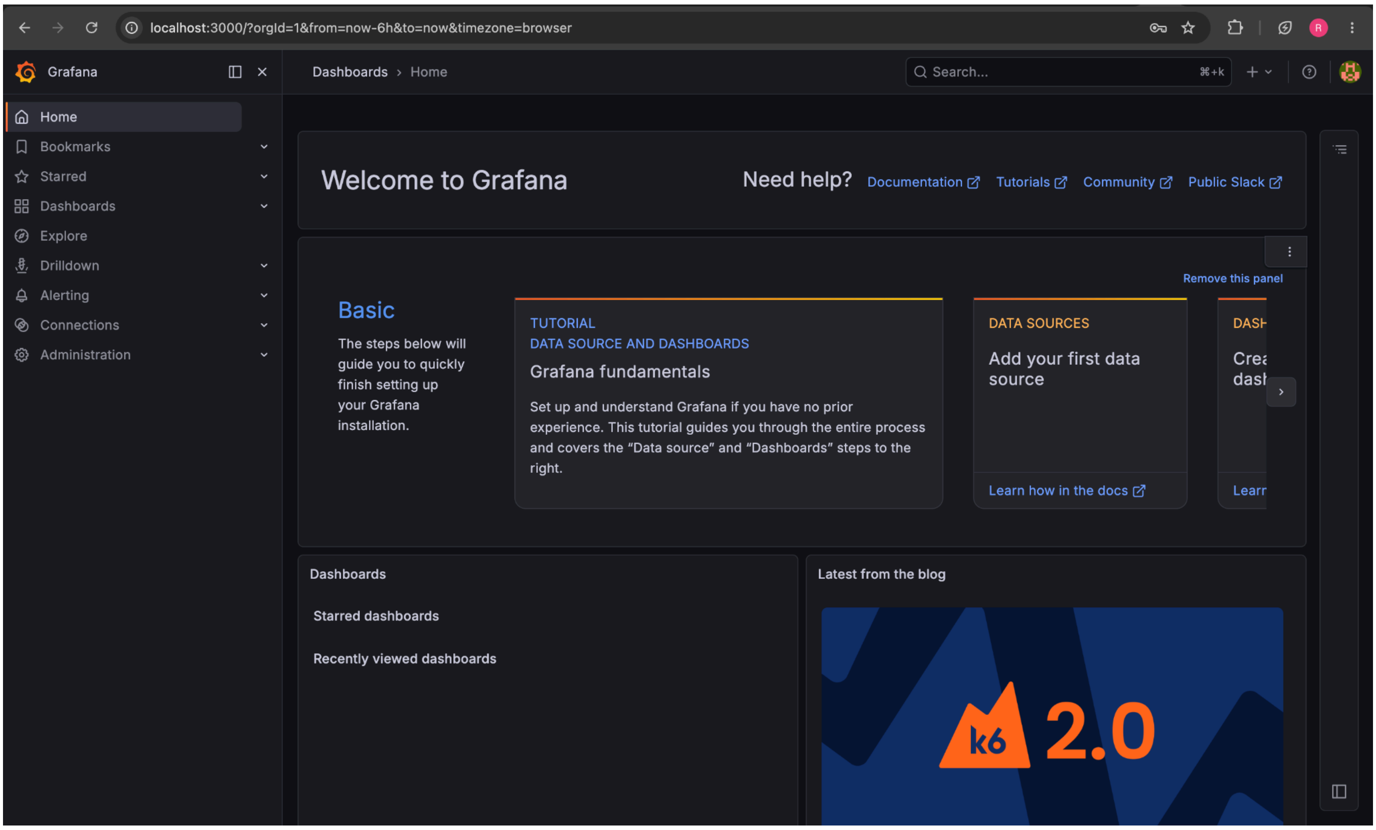
Step 7 :- Prometheus monitors metrics.

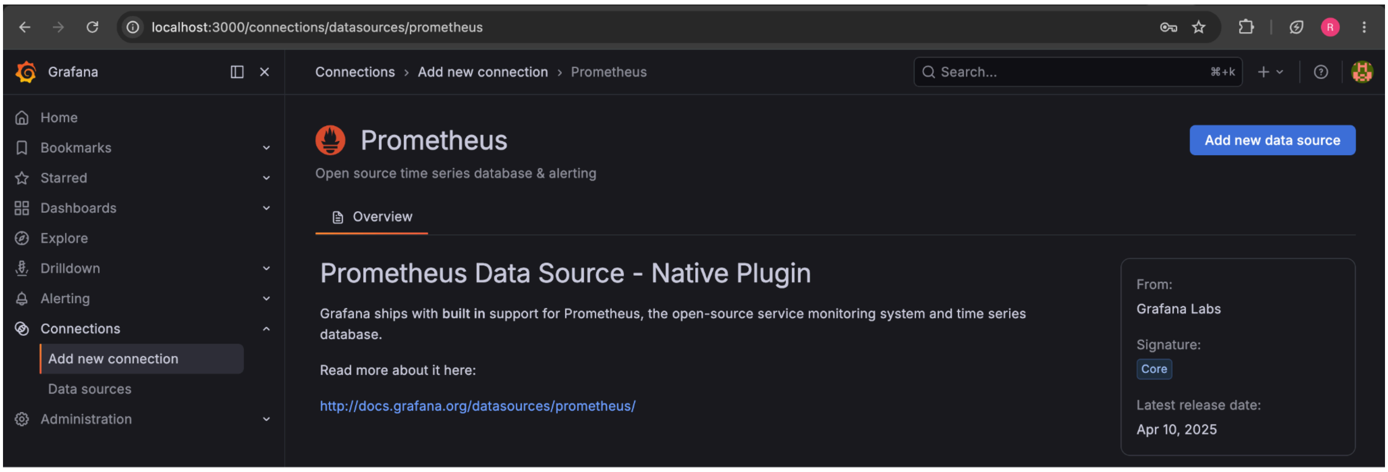


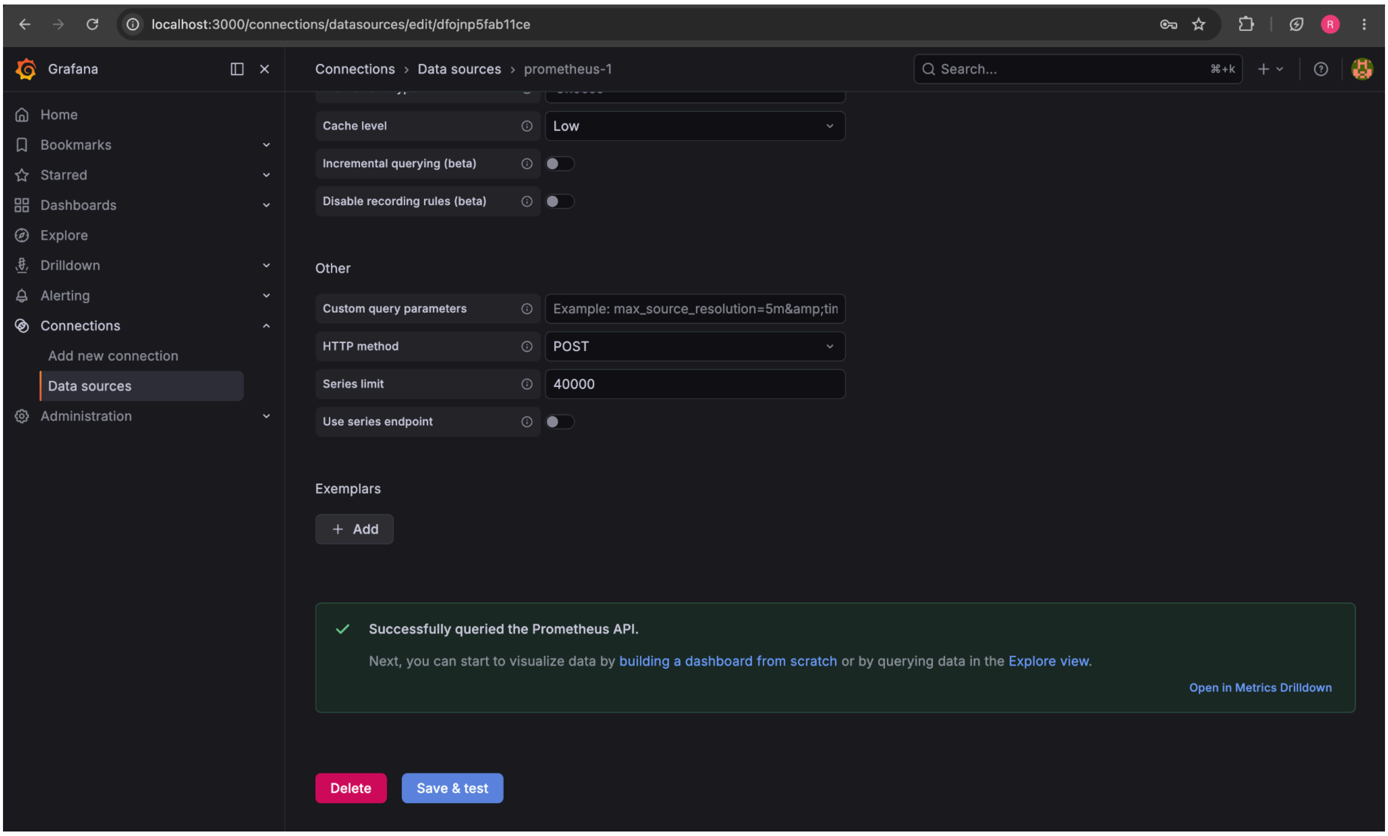


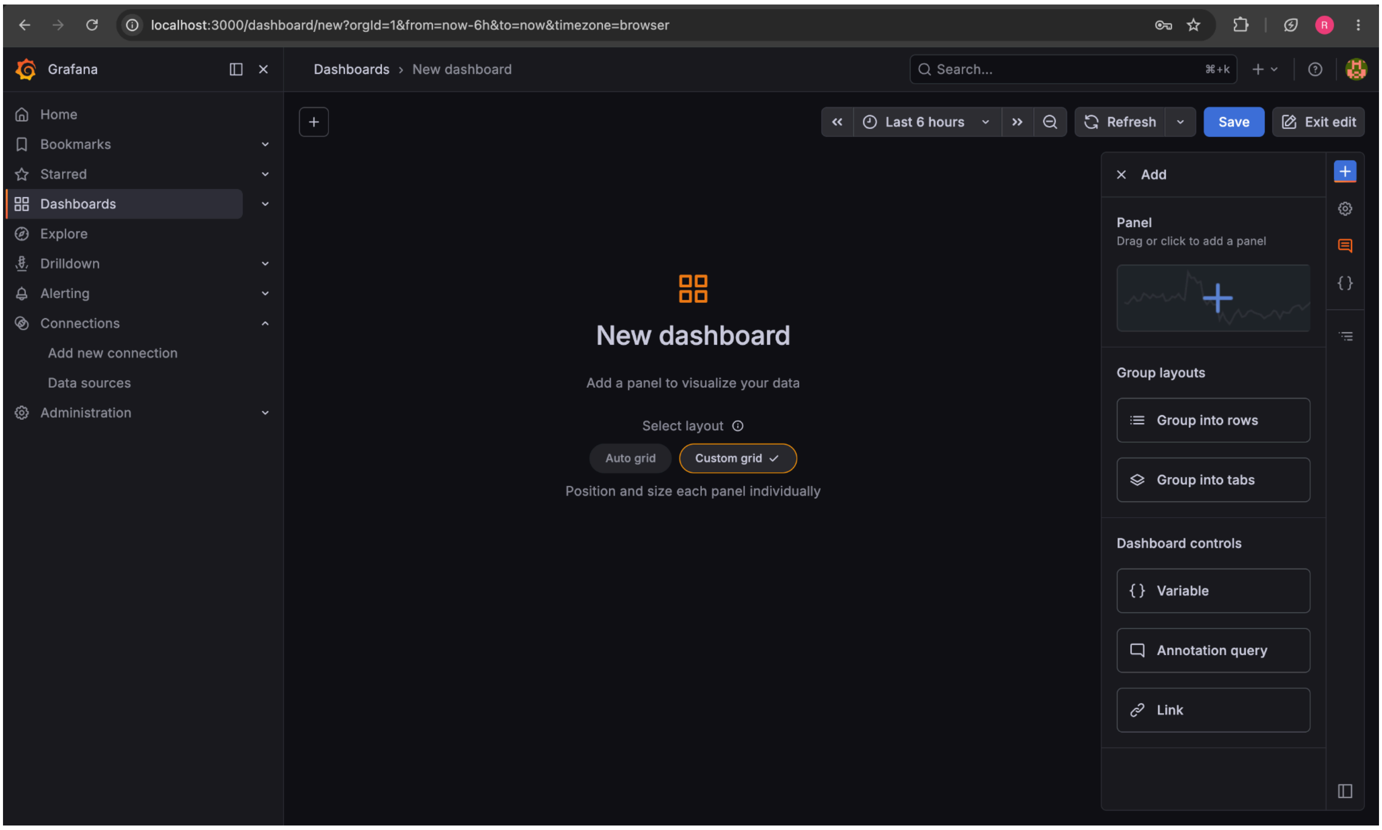
Step 8 :- Grafana visualizes metrics.

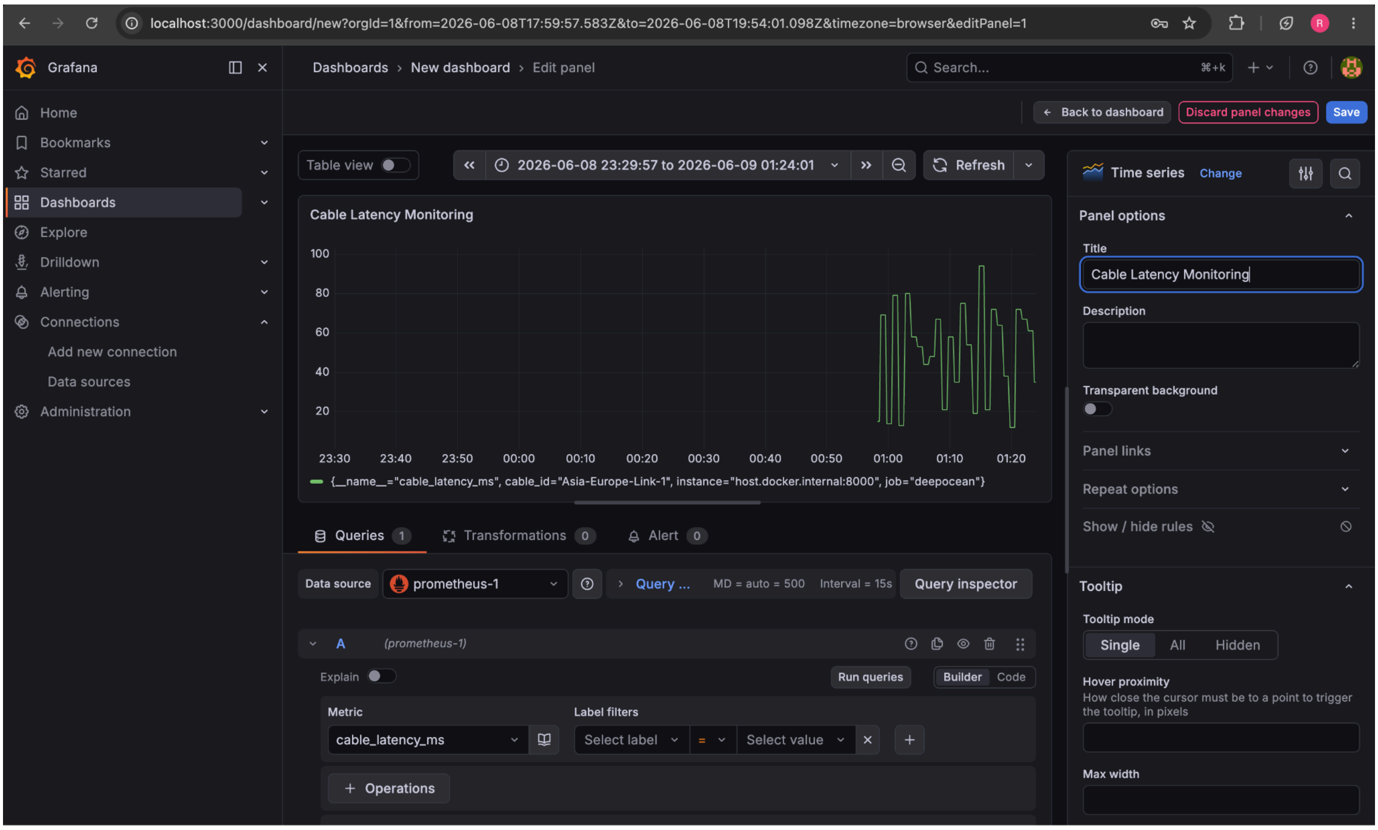

localhost:3000/login
Welcome to Grafana
Email or username
email or username
Password
password
Log in
Forgot your password?

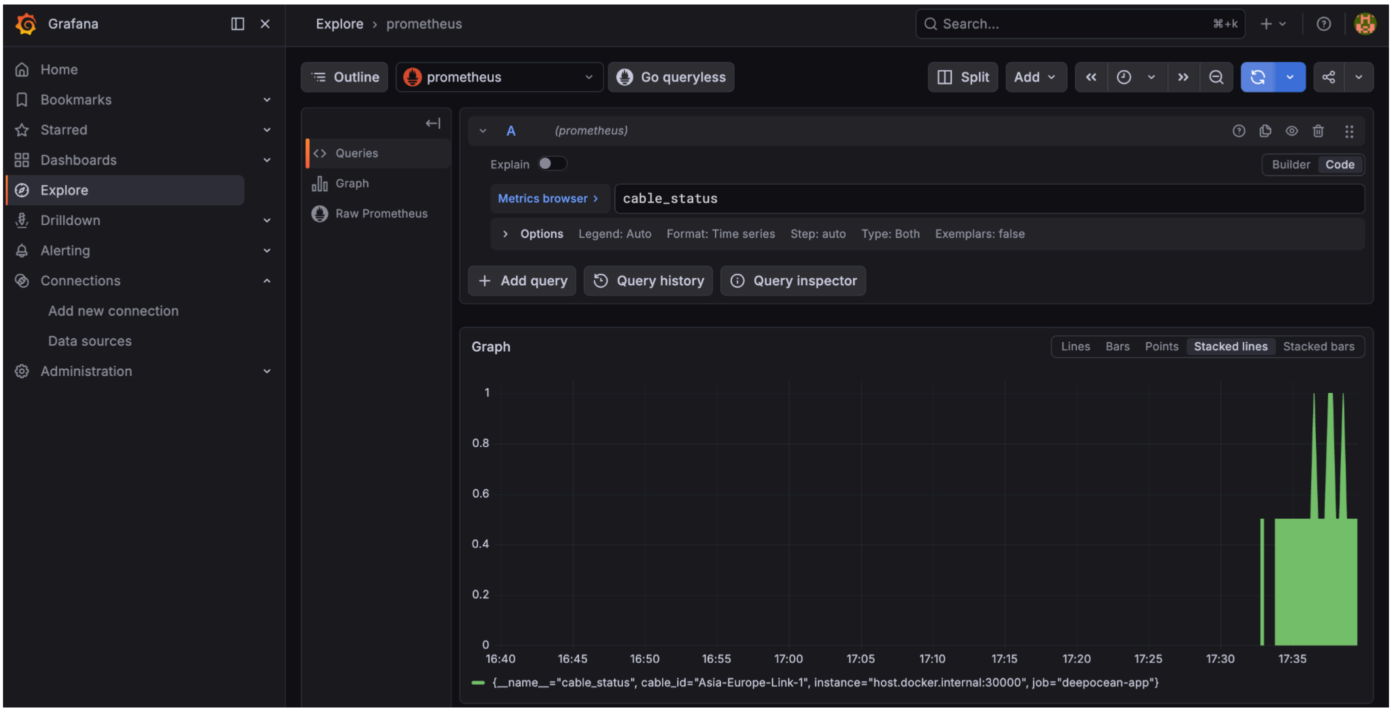

localhost:3000/?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Dashboards › Home
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Administration
Welcome to Grafana
Need help?
Documentation
Tutorials
Community
Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
k6
2.0

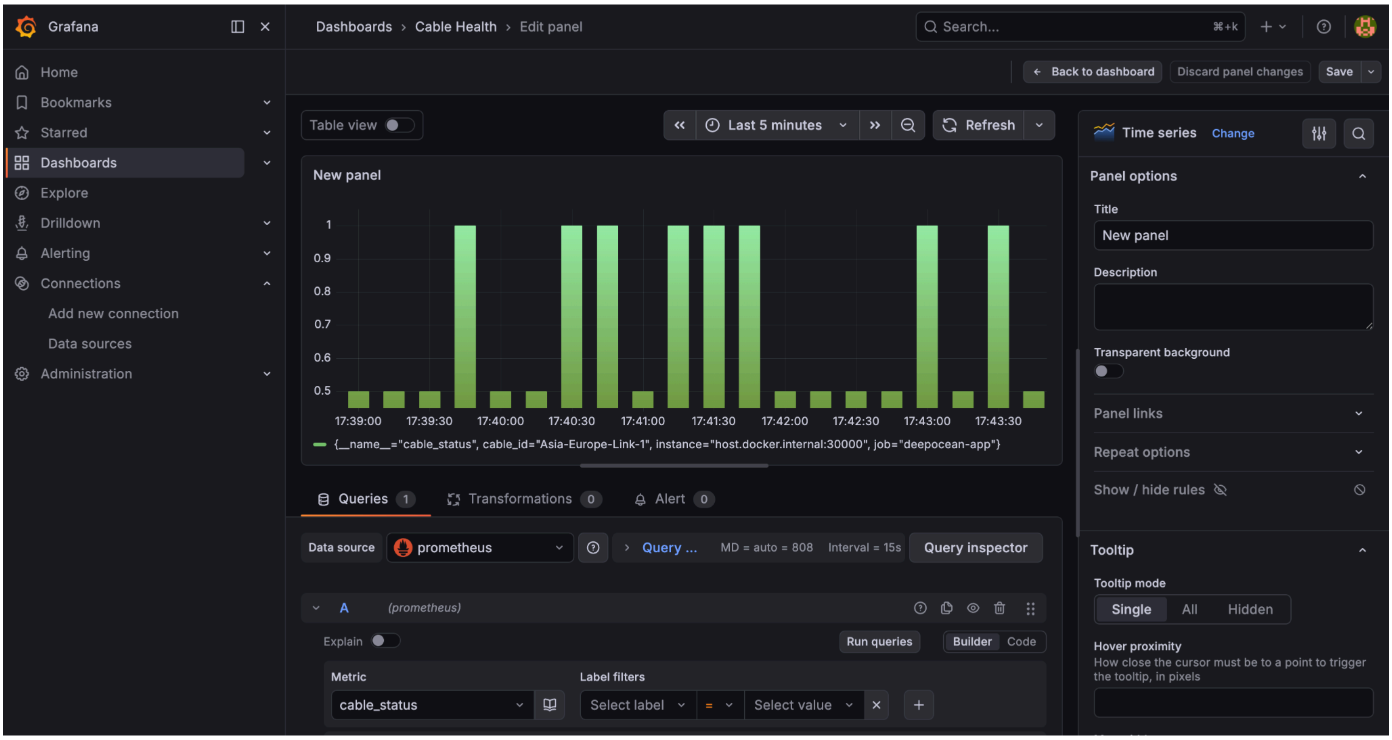

localhost:3000/connections/datasources/prometheus
Grafana
Connections › Add new connection › Prometheus
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Prometheus
Open source time series database & alerting
Add new data source
Overview
Prometheus Data Source - Native Plugin
Grafana ships with built in support for Prometheus, the open-source service monitoring system and time series database.
Read more about it here:
http://docs.grafana.org/datasources/prometheus/
From:
Grafana Labs
Signature:
Core
Latest release date:
Apr 10, 2025


localhost:3000/connections/datasources/edit/dfojnp5fab11ce
Grafana
Connections › Data sources › prometheus-1
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Cache level
Low
Incremental querying (beta)
Disable recording rules (beta)
Other
Custom query parameters
Example: max_source_resolution=5m&tin
HTTP method
POST
Series limit
40000
Use series endpoint
Exemplars
Add
Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard from scratch or by querying data in the Explore view.
Open in Metrics Drilldown
Delete
Save & test

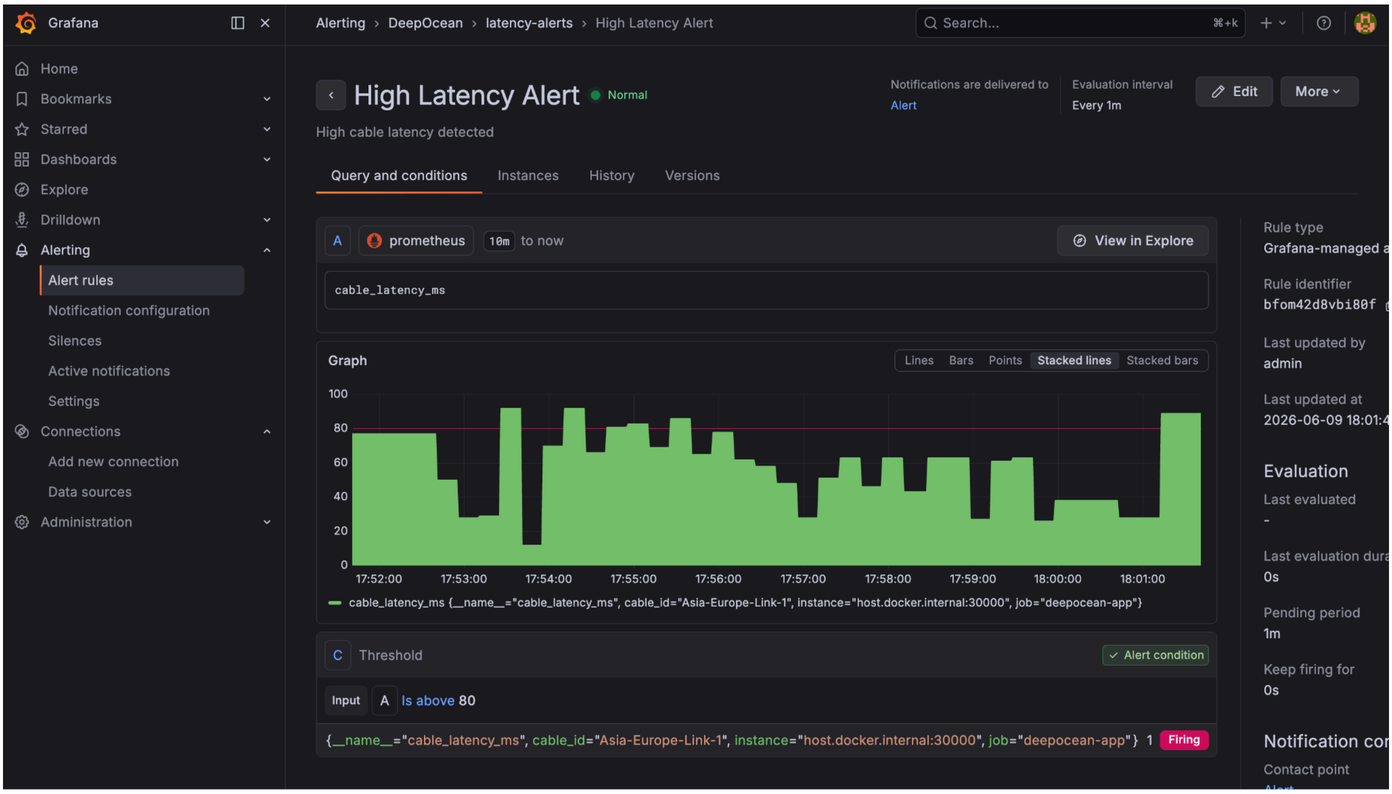

localhost:3000/dashboard/new?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Dashboards › New dashboard
Search...
Last 6 hours
Refresh
Save
Exit edit
New dashboard
Add a panel to visualize your data
Select layout
Auto grid
Custom grid
Position and size each panel individually
Add
Panel
Drag or click to add a panel
Group layouts
Group into rows
Group into tabs
Dashboard controls
Variable
Annotation query
Link

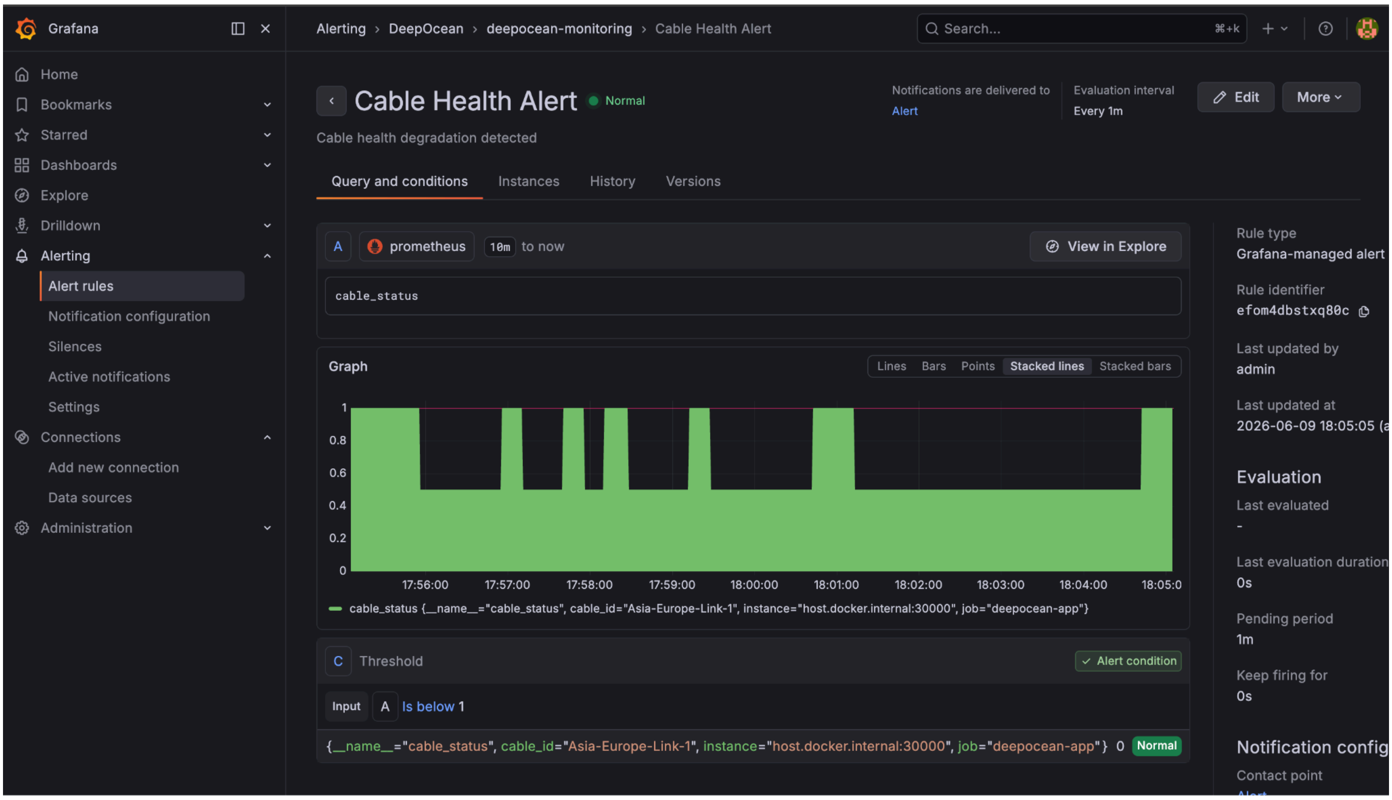

localhost:3000/dashboard/new?orgId=1&from=2026-06-08T17:59:57.583Z&to=2026-06-08T19:54:01.098Z&timezone=browser&editPanel=1
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Dashboards › New dashboard › Edit panel
Search...
Back to dashboard
Discard panel changes
Save
Table view
2026-06-08 23:29:57 to 2026-06-09 01:24:01
Refresh
Cable Latency Monitoring
100
80
60
40
20
23:30
23:40
23:50
00:00
00:10
00:20
00:30
00:40
00:50
01:00
01:10
01:20
{__name__="cable_latency_ms", cable_id="Asia-Europe-Link-1", instance="host.docker.internal:8000", job="deepocean"}
Queries 1
Transformations 0
Alert 0
Data source
prometheus-1
Query ...
MD = auto = 500
Interval = 15s
Query inspector
A
(prometheus-1)
Explain
Run queries
Builder
Code
Metric
cable_latency_ms
Label filters
Select label
=
Select value
Operations
Time series
Change
Panel options
Title
Cable Latency Monitoring
Description
Transparent background
Panel links
Repeat options
Show / hide rules
Tooltip
Tooltip mode
Single
All
Hidden
Hover proximity
How close the cursor must be to a point to trigger the tooltip, in pixels
Max width





Step 9 :- Alerts generated.


Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Alert rules
Notification configuration
Silences
Active notifications
Settings
Connections
Add new connection
Data sources
Administration
Alerting › DeepOcean › latency-alerts › High Latency Alert
Search...
High Latency Alert
Normal
High cable latency detected
Notifications are delivered to
Alert
Evaluation interval
Every 1m
Edit
More
Query and conditions
Instances
History
Versions
A
prometheus
10m to now
View in Explore
cable_latency_ms
Graph
Lines Bars Points Stacked lines Stacked bars
100
80
60
40
20
0
17:52:00 17:53:00 17:54:00 17:55:00 17:56:00 17:57:00 17:58:00 17:59:00 18:00:00 18:01:00
cable_latency_ms {__name__="cable_latency_ms", cable_id="Asia-Europe-Link-1", instance="host.docker.internal:30000", job="deepocean-app"}
C
Threshold
Alert condition
Input
A
Is above 80
{__name__="cable_latency_ms", cable_id="Asia-Europe-Link-1", instance="host.docker.internal:30000", job="deepocean-app"} 1
Firing
Rule type
Grafana-managed a
Rule identifier
bfom42d8vbi80f
Last updated by
admin
Last updated at
2026-06-09 18:01:4
Evaluation
Last evaluated
-
Last evaluation dura
0s
Pending period
1m
Keep firing for
0s
Notification cor
Contact point

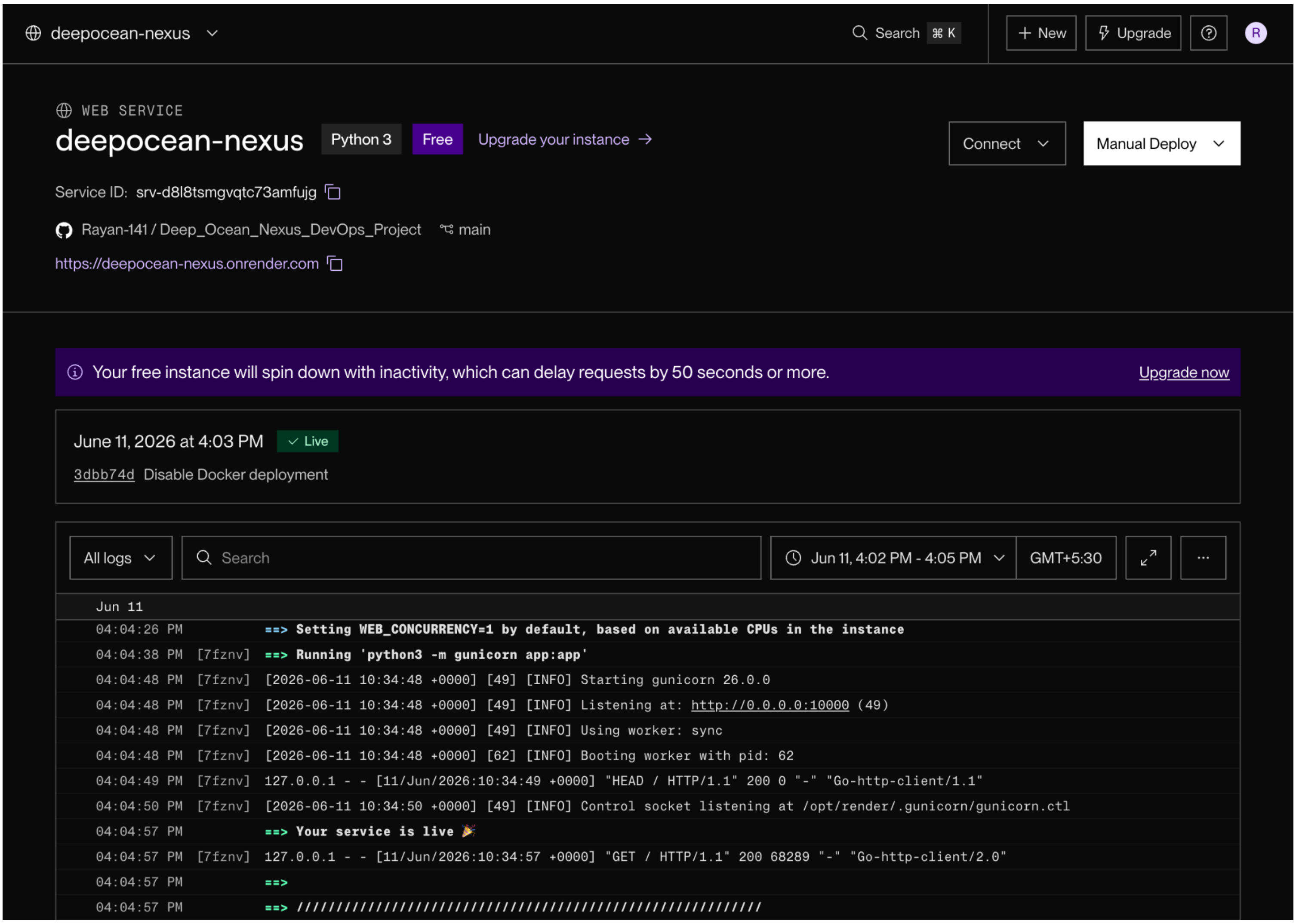

Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Alert rules
Notification configuration
Silences
Active notifications
Settings
Connections
Add new connection
Data sources
Administration
Alerting › DeepOcean › deepocean-monitoring › Cable Health Alert
Search...
Cable Health Alert
Normal
Cable health degradation detected
Notifications are delivered to
Alert
Evaluation interval
Every 1m
Edit
More
Query and conditions
Instances
History
Versions
A
prometheus
10m to now
View in Explore
cable_status
Graph
Lines Bars Points Stacked lines Stacked bars
1
0.8
0.6
0.4
0.2
0
17:56:00 17:57:00 17:58:00 17:59:00 18:00:00 18:01:00 18:02:00 18:03:00 18:04:00 18:05:0
cable_status {__name__="cable_status", cable_id="Asia-Europe-Link-1", instance="host.docker.internal:30000", job="deepocean-app"}
C
Threshold
Alert condition
Input
A
Is below 1
{__name__="cable_status", cable_id="Asia-Europe-Link-1", instance="host.docker.internal:30000", job="deepocean-app"} 0
Normal
Rule type
Grafana-managed alert
Rule identifier
efom4dbstxq80c
Last updated by
admin
Last updated at
2026-06-09 18:05:05 (a
Evaluation
Last evaluated
-
Last evaluation duration
0s
Pending period
1m
Keep firing for
0s
Notification config
Contact point

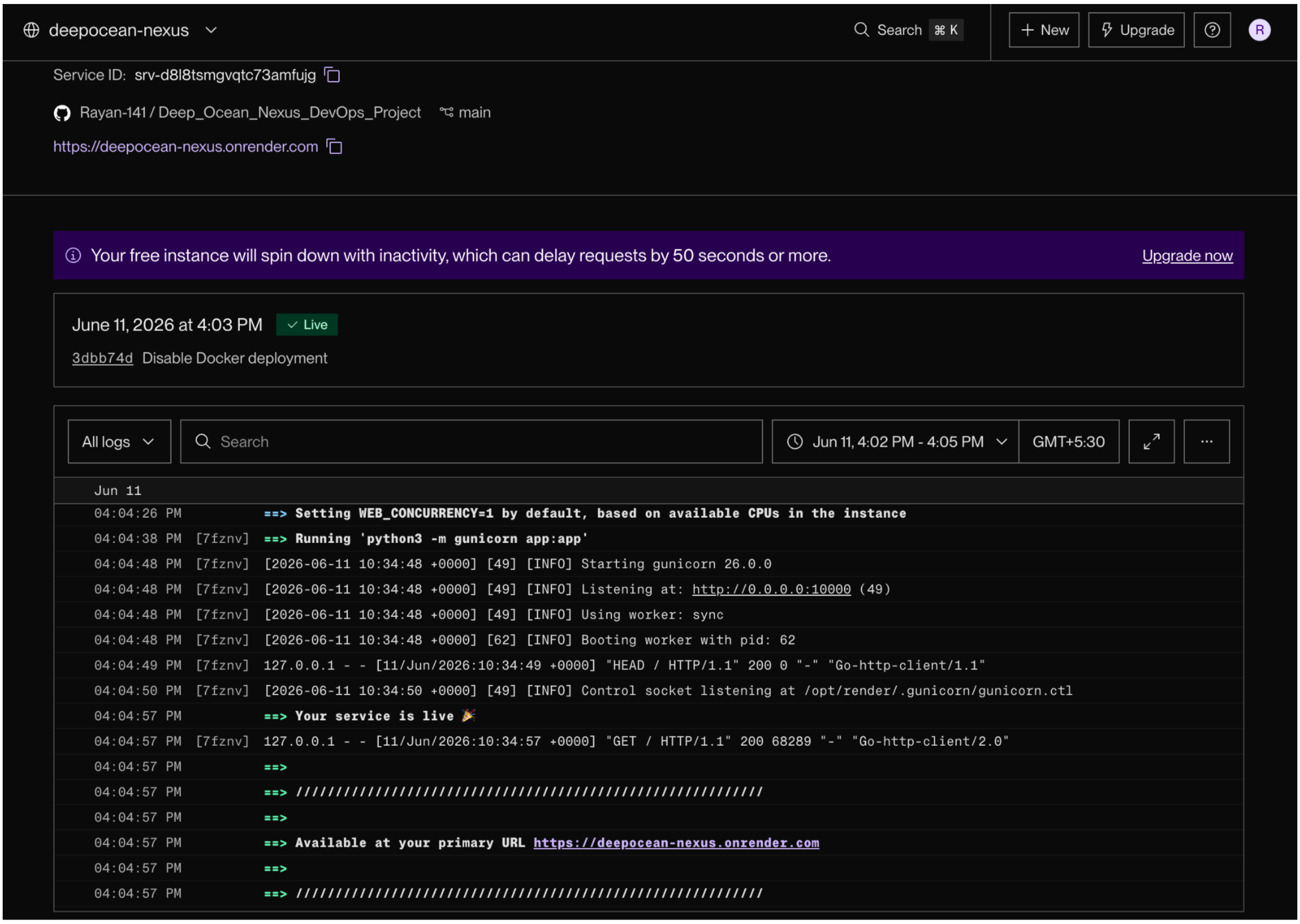
Step 10 :- Backup and recovery available.

```
apple@Rayans-MacBook-Air logging % cd ~/Desktop/DeepOcean_Nexus_DevOps_Project

mkdir -p backup
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % ls
app                 disaster_recovery.md    logging             project_report.md       screenshots
architecture.md     jenkins                 logging.md          project_tracker.txt     terraform
backup              kubernetes              monitoring          README.md               viva_questions.md
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % kubectl get all -n deepocean -o yaml > backup/deepocean-backup.yaml
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % ls backup
deepocean-backup.yaml
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % wc -l backup/deepocean-backup.yaml
    1015 backup/deepocean-backup.yaml
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % kubectl get all -n deepocean
NAME                                 READY   STATUS    RESTARTS   AGE
pod/deepocean-app-7684bfccf9-fzj6d   1/1     Running   0          65m
pod/deepocean-app-7684bfccf9-h5b5j   1/1     Running   0          65m
pod/deepocean-app-7684bfccf9-v6hpq   1/1     Running   0          65m

NAME                        TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/deepocean-service   NodePort   10.105.165.192   <none>        5000:30000/TCP   17h

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/deepocean-app   3/3     3            3           17h

NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/deepocean-app-5ddfffd555   0         0         0       17h
replicaset.apps/deepocean-app-7684bfccf9   3         3         3       65m
replicaset.apps/deepocean-app-794595cf87   0         0         0       5h43m
replicaset.apps/deepocean-app-857494df9c   0         0         0       5h33m

NAME                                                REFERENCE                  TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/deepocean-app   Deployment/deepocean-app   cpu: 14%/50%   3         10        3          11m
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
```

```
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % kubectl get all -n deepocean
NAME                                 READY   STATUS    RESTARTS   AGE
pod/deepocean-app-7684bfccf9-9gptx   1/1     Running   0          107s
pod/deepocean-app-7684bfccf9-mqr6h   1/1     Running   0          107s
pod/deepocean-app-7684bfccf9-s54lc   1/1     Running   0          107s

NAME                        TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/deepocean-service   NodePort   10.105.165.192   <none>        5000:30000/TCP   17h

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/deepocean-app   3/3     3            3           107s

NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/deepocean-app-7684bfccf9   3         3         3       107s

NAME                                                REFERENCE                  TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/deepocean-app   Deployment/deepocean-app   cpu: 12%/50%   3         10        3          15m
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project % curl http://localhost:30000/health
{"service":"DeepOcean Nexus","status":"healthy"}
apple@Rayans-MacBook-Air DeepOcean_Nexus_DevOps_Project %
```

Live Deployment:-





Platform: Render Cloud

Live URL: https://deepocean-nexus.onrender.com

Benefits:

- Public access
- Continuous deployment
- Cloud hosting

Conclusion:-

DeepOcean Nexus successfully demonstrates a modern cloud-native DevOps ecosystem capable of supporting critical submarine communication infrastructure through automation, monitoring, centralized logging, disaster recovery, and scalable deployment practices.